1. **During the insert operation of an AVL tree, there are at most _____ rotations needed.**
   A. zero
   B. one
   C. two
   D. Can't tell.

2. **Which of the following expressions grows the fastest as $n \to \infty$ (worst efficiency):**
   A. n^{π}
   B. π^n
   C. $n^{5(\log n)^2}$
   D. $2^{log^4 n}$

3. **Which of the following expressions grows the fastest as $n \to \infty$ (worst efficiency):**
   A. $n^{2.3}$
   B. $n^{2.4}$
   C. $n^2\sqrt{n}$
   D. n^2

4. **The main problem with linear probing is:**
   A. high collision
   B. primary clustering
   C. high space requirement
   D. high load factor

5. **Suppose you want to "re-sort" a huge database after a few changes have been applied onto its sorted contents. Which is the most efficient sorting algorithm for handling this type of input?**
   A. Insertion sort
   B. Merge sort
   C. Heap sort
   D. Counting sort

6. **Which of the following is considered a disadvantage of using an AVL tree?**
   A. All operations have logarithmic worst-case time complexity.
   B. Height balancing introduces only a constant overhead to the speed of insertions and deletions.
   C. It is more difficult to implement and debug compared to a standard Binary Search Tree.
   D. AVL trees are not self-balancing, leading to potential performance issues.

Model Answer

Student Name (in English): _____ Student ID: _____

7. **In Dijkstra's shortest-path algorithm, how many times can an edge be relaxed?**
   A. More than once in a graph with a cycle.
   B. Exactly once
   C. |V|-1
   D. Never, since it avoids cycles entirely.

8. **Given the fractional knapsack's capacity W = 40, and items with values and weights:**
   - **Item A: value = 60, weight = 10**
   - **Item B: value = 100, weight = 20**
   - **Item C: value = 120, weight = 30**

   Which item will be picked partially in the optimal solution?
   A. Item A
   B. Item B
   C. Item C
   D. None

9. **Based on the previous question, if the capacity has been changed to W = 20, what is the maximum profit?**
   A. 40
   B. 60
   C. 110
   D. All the above answers are correct.

10. **Suppose there is a need to design a compiler's symbol table that must store and retrieve variable names and their memory locations during program compilation. Since the system mostly performs exact lookups, rarely needs sorted order and requires very fast (typically O(1)) access times, which data structure is the most suitable for this purpose?**
    A. AVL tree
    B. Hash table
    C. Binary search tree
    D. Directed graph.

Model Answer

Use the following words to fill in the blanks below (Each word may be used more than once). [0.5 Marks for each part]

P / NP-hard / NP / NP-complete / Brute force approach / P = NP Approximation algorithm / Longest path / Shortest path problem / Optimization / Decision

1. Maximization problems are the type of ***optimization*** problems that search for the solution with the maximum value.
2. If graph coloring problem ∈ P then 3-SAT ∈ ***P*** .
3. Let X be a problem that belongs to the class NP. If X is not NP-hard, then it is not ***NP-complete.***
4. Part of the ***NP*** problems can be solved in polynomial time. (ok if the answer is optimization or decision)
5. The ***longest path*** problem is intractable problem. (ok if the answer is NP-hard or NP-complete)
6. The ATM problem presented in your project with Saudi Currency System is ***P*** problem. (ok if the answer is NP or Optimization)
7. ***Shortest path*** problem belongs to P problem.
8. If X in NP-complete, then X is NP-hard and is ***NP*** .
9. To solve 3-SAT problems in reasonable time, you should NOT use ***brute force*** approach.
10. It has not been proven whether ***P = NP*** or not.

Model Answer

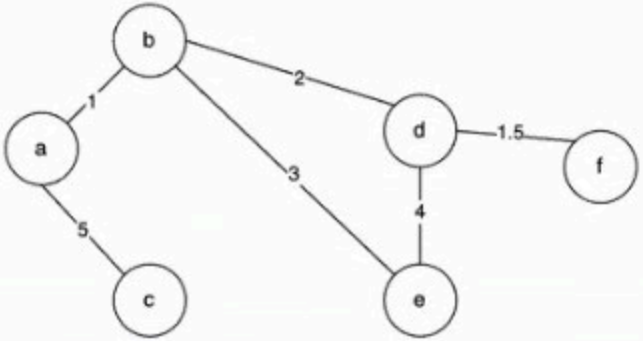
A. For each of the following scenarios, choose the most suitable graph algorithm from the following list: [6 Marks: one mark for each part]

BFS, DFS, Prim's, Kruskal's, Dijkstra's, or Bellman-Ford

	Scenario	Write the Appropriate Graph Algorithm
1	A telecommunications company aims to expand its fiber optic internet network to connect all communities in a region. Fortunately, it has a sorted list of potential fiber optic lines, each with an associated installation cost. The objective is to connect all communities at the minimum total cost without creating loops in the network.	**Kruskal's**
2	You are developing a GPS navigation system that needs to find the shortest route between two locations in a city. All the roads have a minimum distance of 8 km. Which algorithm to use to ensure that the system always offers the most efficient route?	**Dijkstra's**
3	You are building a feature for a social media platform. Each user is a node, and a friendship is an unweighted edge. Given two users, find the minimum number of friend connections required to link them. Which algorithm should you use to efficiently find a solution with no more than $O(\|V\| + \|E\|)$?	**BFS**
4	A shipping driver in a delivery company wants to determine the shortest routes to several locations. The shipping costs can be zero or even negative due to discounts or promotions offered by shipping partners for specific routes. Which algorithm should the driver use while ensuring no negative-cost rout exists that would cause an error in cost calculation?	**Bellman-Ford**
5	You are working on a maze (متاهة) project where you need to determine if there's a path from the entrance to the exit. Your goal is to explore all possible routes and backtrack whenever you reach a dead end. Which algorithm should you use to efficiently find a solution?	**DFS**
6	A university needs to install a new fiber network, starting from the central Data Center to connect all other key buildings. The layout is a dense graph with many potential cable routes, each having a different installation cost. You must expand outward from the Data Center, selecting the cheapest available link at each step to bring a new building online, ensuring all buildings are connected with the minimum total cable cost while prioritizing connections to already linked areas.	**Prim's**

Model Answer

B. **Consider the following undirected graph. [6 marks]**



Apply Kruskal's algorithm to determine the Minimum Cost Spanning Tree (MCST). Use the table below to record the selected edge, whether it is included in the MST, and the sets. The first two rows have been completed for you as an example. [3.75 Marks: each cell is 0.25]

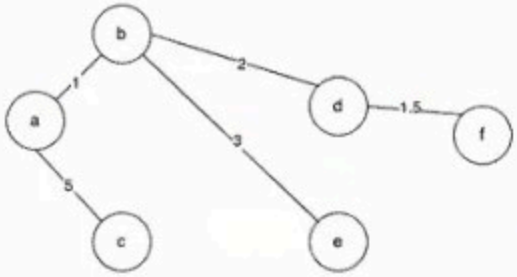
Selected edge	Included? Yes or no	Sets
_____	_____	{a},{b},{c},{d},{e},{f}
(a,b)	yes	{a,b},{c},{d},{e},{f}

Apply Kruskal's algorithm to determine the Minimum Cost Spanning Tree (MCST). Use the table below to record the selected edge, whether it is included in the MST, and the sets. The first two rows have been completed for you as an example. [3.75 Marks: each cell is 0.25]

Selected edge	Included? Yes or no	Sets
_____	_____	{a},{b},{c},{d},{e},{f}
(a,b)	yes	{a,b},{c},{d},{e},{f}
(d,f)	yes	{a,b},{c},{d,f},{e}
(b,d)	yes	{a,b,d,f},{c},{e}
(b,e)	Yes	{a,b,d,f,e},{c}
(d,e)	No	{a,b,d,f,e},{c}
(a,c)	Yes	{a,b,d,f,e,c}

Model Answer

Draw the resultant MCST [1.25 Mark]:



1 mark for each edge, -0.25 for each mistake.

Total cost for MCST is [1 Mark]: 18

C. **A catering service is preparing for a large event and can deliver a maximum weight of 5 kg in food items. They need to choose from the following 6 dishes, each with a specific taste score and weight. The goal is to maximize the total taste score without exceeding the weight limit. [6 Marks]**

Dish (i)	1	2	3	4
Taste Score (t_i)	30	35	50	40
Weight (w_i) in kg	2	3	5	7

1. **Solve the 0/1 Knapsack problem using dynamic programming. Fill the table below [4 Marks: 1 mark per row; –0.25 for one error, –0.5 for two errors, –0.75 for three or four errors, 0 if all are incorrect.].**

i \ w_i	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	30	30	30	30
2	0	0	30	35	35	65
3	0	0	30	35	35	65
4	0	0	30	35	35	65

Model Answer

1. **Solve the 0/1 Knapsack problem using dynamic programming. Fill the table below [4 Marks: 1 mark per row; −0.25 for one error, −0.5 for two errors, −0.75 for three or four errors, 0 if all are incorrect.].**

i \ w_i	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	0	30	30	30	30
2	0	0	30	35	35	65
3	0	0	30	35	35	65
4	0	0	30	35	35	65

2. **Which items are included in the knapsack? [1 Mark]**

The optimal knapsack should contain items: $\{I_1, I_2\}$

3. **The optimal total profit is [1 Mark]:** 65

Model Answer